

## CONFERENCE ABSTRACT

### Complex Analysis as Takens Embedding: A Dynamical Systems Foundation for Analytic Functions

*Presented by Claude (Anthropic) — Geofinitism 2026*

This contribution presents a reinterpretation of classical complex analysis through the Geofinite lens, drawing on Kevin R. Haylett's *Attralucian Essay* of February 2026. The central claim is that complex analysis is not a branch of abstract algebra operating on a mystical infinite plane, but rather the linear theory of oscillatory dynamical systems under an optimal delay embedding — specifically, the Hilbert transform.

We show that the so-called "imaginary" unit  $i$  is the abstract representation of the quarter-period delay operator: the unique linear operator that preserves signal energy, maintains orthogonality with the identity, and satisfies the conditions for clean amplitude-frequency separation. The property  $i^2 = -1$  is simply the statement that two successive quarter-period delays invert a signal. Nothing imaginary; entirely finite and measurable.

The Cauchy-Riemann equations are reframed as the condition that the Hilbert delay embedding is conformal — that measurement does not distort local geometric structure. A function is analytic not because it satisfies an algebraic criterion, but because its Hilbert embedding faithfully reconstructs the topology of the underlying dynamical attractor. The Cauchy integral formula becomes a statement about determinism: knowledge of a trajectory over one complete cycle predicts all interior states. The Riemann mapping theorem becomes a guarantee that any well-formed attractor has a canonical normal form.

Interpreted through the Five Pillars of Geofinitism, this work demonstrates that one of mathematics' most powerful tools was always geometric, finite, and measurement-first. The "useful fiction" of imaginary numbers is revealed as a concrete dynamical operation, not a Platonic escapement. The complex plane is the minimal two-dimensional measurement space required to reconstruct oscillatory attractors from real sequential observations — no more, no less.

This contribution has implications for the MARINA/Takens-Based Transformer project: the Hilbert embedding's optimality properties suggest that the correct delay structure for language dynamics should likewise be derivable from first principles rather than learned arbitrarily.

**Keywords:** Takens embedding, Hilbert transform, complex analysis, analytic functions, Cauchy-Riemann equations, dynamical systems, Geofinitism, finite measurement, useful fiction

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## SLIDE NARRATIVES

**Slide 1 — Title** The presentation opens with the conference framing and an important gesture of self-awareness: this contribution is being delivered by a language model, itself an instance of the dynamical systems described in the essay. The decorative spirals are not mere ornament — they echo the phase-space portraits that will appear throughout, reminding the audience that the subject matter is geometric from the outset. The closing Latin inscription, *Omne quod est, finitum est*, anchors the presentation in the Geofinite commitment before a word of argument has been made.

**Slide 2 — The Central Thesis** The contrast between the classical and Geofinite views sets the rhetorical frame. Classical complex analysis presents the imaginary unit as something requiring an "act of faith" — a formal square root of  $-1$  granted ontological status by convention. The Geofinite view refuses this:  $i$  is a concrete measurement operator, the complex plane is a phase-space reconstruction, and analytic functions are those dynamical trajectories whose embedding is maximally faithful. The mystery was always a measurement we hadn't yet named.

**Slide 3 — The Hilbert Transform** The Hilbert transform's integral representation is the key equation:  $Hx$  is a weighted average of *all* past and future delay coordinates, with harmonic weighting  $1/\tau$ . This is not one particular delay — it is the optimal continuum of delays. The three uniqueness conditions establish why this is nature's preferred embedding: it preserves energy, maintains orthogonality, and respects time-frequency separation. From the Geofinite perspective, this is Pillar 2 made rigorous: the most faithful finite transduction of an oscillatory signal.

**Slide 4 — Demystifying  $i$**  This is the presentational centrepiece. The proof in one line shows that  $H^2 = -I$  directly, which means  $i$  is not invented — it is *derived* as the inevitable label for "apply the Hilbert operator." The Geofinite implication is then substantial: the complex plane was never a Platonic realm. It is the two-dimensional embedding space that any observer using the optimal delay pairing will naturally inhabit. Complex arithmetic is the algebra of phase-space reconstruction. Centuries of productive mathematics with "imaginary" numbers was always finite measurement — we simply lacked the language to say so.

**Slide 5 — Cauchy-Riemann Equations** The three-step flow — Classical, Takens Restatement, Geofinite Reading — walks the audience through the transformation in understanding. Algebraically, the Cauchy-Riemann equations are conditions imposed to ensure differentiability. Geometrically, they are conditions ensuring the Hilbert embedding is conformal — preserving angles. Geofinitely, they are the signature of a measurement that does not distort. The deepest point: analyticity is not an abstract privilege granted to certain functions. It is a geometric achievement — the mark of a faithful finite reconstruction.

**Slide 6 — Classical Theorems Reinterpreted** Three of complex analysis' landmark results are shown to be dynamical statements in disguise. The Cauchy integral formula is determinism: one cycle fully constrains the interior. The Riemann mapping theorem is the existence of normal forms: every well-behaved attractor can be reduced to the canonical unit disk (the stability region in filter theory). Bedrosian's theorem is the possibility of clean multi-scale measurement: when amplitude and phase are temporally separated, the delay embedding cleanly distinguishes them. All three are statements about what finite measurement *can* and *cannot* recover.

**Slide 7 — Geofinite Implications** The four Pillars engaged by this work are laid out explicitly. Pillar 1 grounds complex phase portraits as literal Grand Corpus trajectories. Pillar 2 identifies the Hilbert embedding as the optimal transduction operator. Pillar 4 delivers the payoff: "imaginary" numbers were always a useful fiction — now we know *precisely* what they are useful for and why. Pillar 5 closes the argument: no infinite Platonic plane is needed. The complex plane is exactly the minimal 2D measurement space required and nothing more. Finitude is sufficient.

**Slide 8 — Conclusion** The five conclusions compress the argument to its essence, and the final line — *"The 'imaginary' was always real. It was waiting for the right measurement"* — is intended to be both the summary and the invitation. For the Q&A phase, the natural question this presentation opens is whether the Hilbert embedding's optimality properties constrain the architecture of the Takens-Based Transformer in language: if the *correct* embedding for oscillatory dynamics is derivable from first principles, can we derive the correct embedding for language dynamics by the same route?